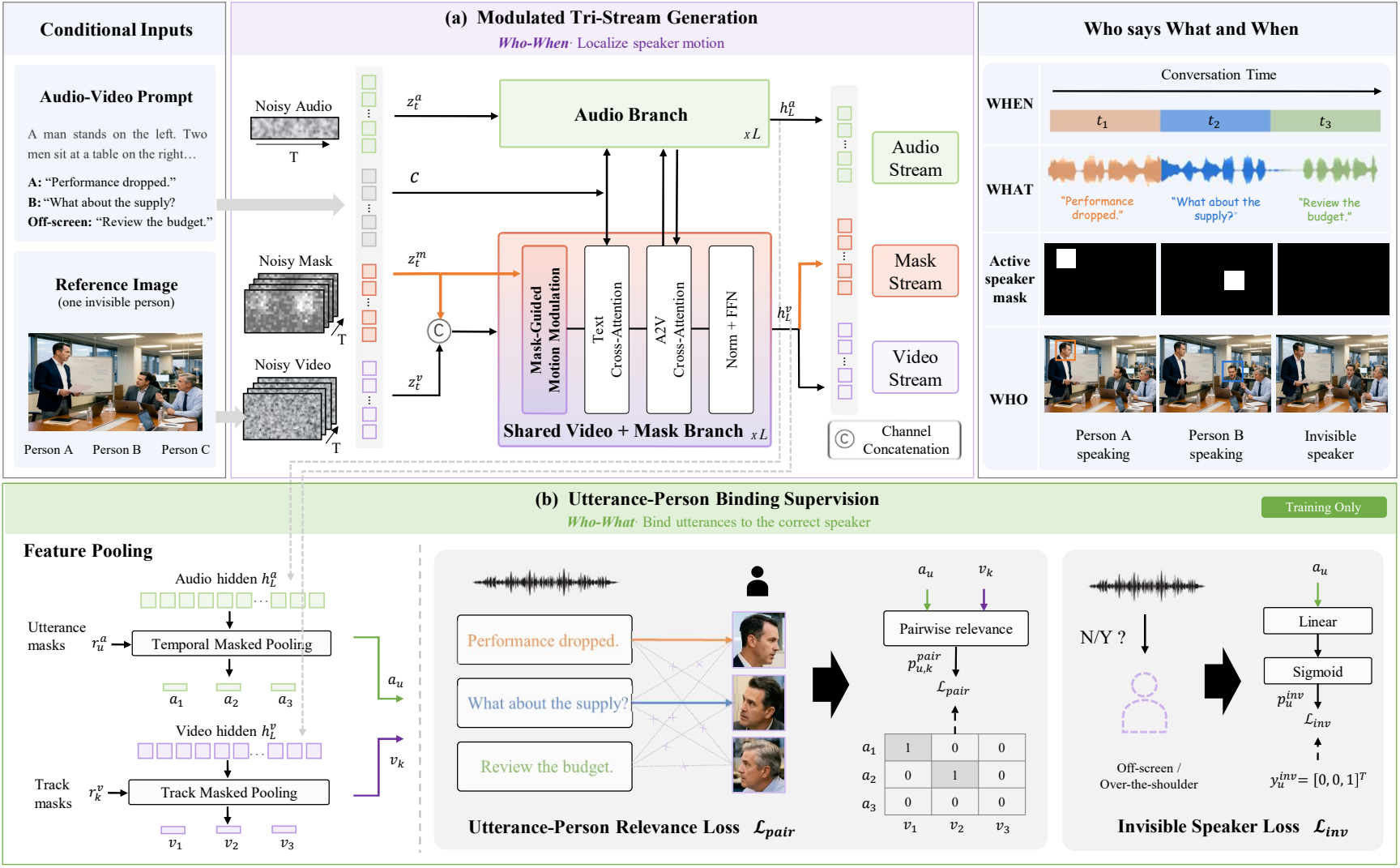

Conditional Inputs
Audio-Video Prompt
A man stands on the left. Two men sit at a table on the right…
A: "Performance dropped."
B: "What about the supply?
Off-screen: "Review the budget."
Reference Image
(one invisible person)
Person A
Person B
Person C
(a) Modulated Tri-Stream Generation
Who-When· Localize speaker motion
Noisy Audio
Noisy Mask
Noisy Video
T
Audio Branch
x L
Mask-Guided Motion Modulation
Text Cross-Attention
A2V Cross-Attention
Norm + FFN
Shared Video + Mask Branch x L
Audio Stream
Mask Stream
Video Stream
Channel Concatenation
Who says What and When
WHEN
Conversation Time
WHAT
"Performance dropped."
"What about the supply?"
"Review the budget."
Active speaker mask
WHO
Person A speaking
Person B speaking
Invisible speaker
(b) Utterance-Person Binding Supervision
Who-What· Bind utterances to the correct speaker
Training Only
Feature Pooling
Audio hidden
Utterance masks
Temporal Masked Pooling
Video hidden
Track masks
Track Masked Pooling
Performance dropped.
What about the supply?
Review the budget.
Pairwise relevance
Utterance-Person Relevance Loss
N/Y ?
Off-screen / Over-the-shoulder
Linear
Sigmoid
Invisible Speaker Loss